

Assignment

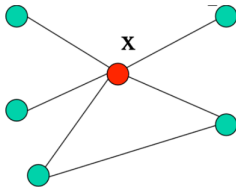
1. In chapter 4, we introduced network generation models includes Erdos-Renyi, Watts-Strogatz, and Barabasi-Albert.

- Please write code to implement any one of the network model and generate a simulation network with size 30.
- Then visualise the generated network by Cytoscape.

Please submit code, the screenshot of the network visualization, and the file of adjacency matrix (or adjacency list).

2. Given the following network,

- what is the degree and betweenness for x?
- What is characteristic path length for the network?



3. According to the CLR (Context Likelihood of Relatedness) process, given the gene expression profile, we have computed the MI matrix (mutual information for each gene pair ij)

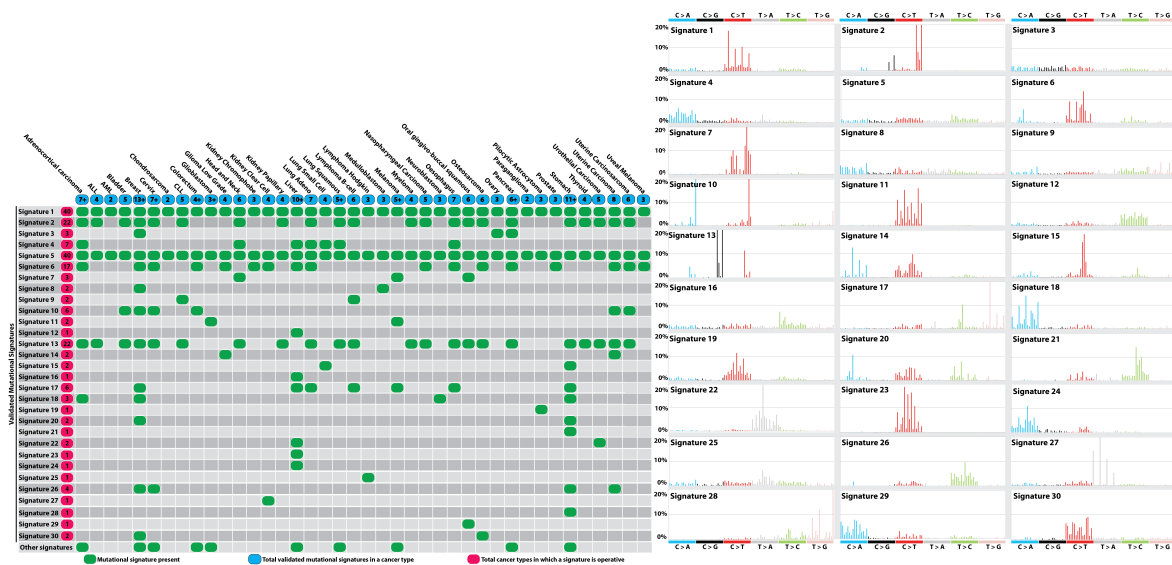
- Please compute the Zi matrix (positive z-score for MI_{ij} respect to row) and Zj matrix(positive z-score for MI_{ij} respect to column). You may write code to finish these steps (optional).
- Please calculate the CLR matrix and list the top 3 gene-gene interactions.

expression

	G1	G2	G3	G4
Sample1	4	3	3	2
Sample2	5	4	2	3
Sample3	3	2	3	2
Sample4	6	6	5	1
Sample5	4	2	3	1
Sample6	4	5	5	2

MI	G1	G2	G3	G4
G1	1.24	1.01	0.69	0.69
G2	1.01	1.56	1.01	0.78
G3	0.69	1.01	1.01	0.46
G4	0.69	0.78	0.46	1.01

4. Below are COSMIC cancer mutation signatures detected based on NMF (https://cancer.sanger.ac.uk/signatures/signatures_v2/). Consider $X=W*H$, X is mutation profile across cancer samples. After apply NMF to mutation profile, X is decomposed to W (mutation signature) and H (the contribution of different signatures in each cancer types). In the following figure of H (left subfigure) and W (right subfigure).



- What is the major mutation signature types contribute to AML?
- What is the major mutation type (such as C>A, C>G, C>T, T>A, T>C, T>G) related to AML?
- Which mutation signature are domain by T>A (more than 10%)? Which cancer types are highly associated with it?
- We decide to develop a model to diagnose patient. Suppose we have already know the disease status for 1000 patients, and for each patient we also collect the blood(or other tissue if applicable) samples, and did DNA sequencing (get the personal genome). By comparing each personal genome to reference genome, we can get the patient's mutation profiles. Please design a model to do classification, so that for a new patient with the personal genome, while we do not know the disease status, we can predict which cancer type it may be related to. Describe why the model fit this task. [open question]